between reasoning length and correctness: while overly brief responses often lack sufficient reasoning to produce correct answers, excessively long outputs tend to be detrimental to both accuracy and efficiency. These results point to the existence of an optimal reasoning length within the sample distribution—neither too short nor too long—for achieving the best accuracy while maintaining efficiency.

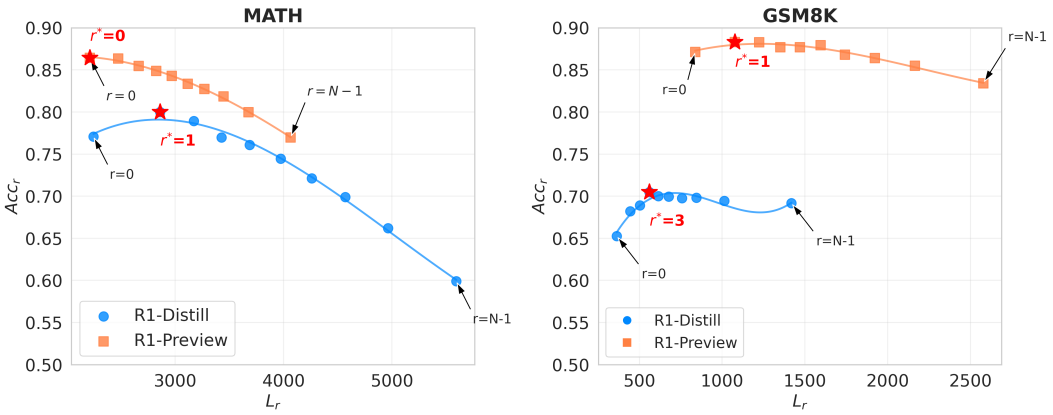

Figure 1: Reasoning length L_r of the r-th shortest response v.s. accuracy Acc_r ($N = 10$). The red marker r^* denotes the rank of responses with the highest accuracy. Results for R1-Distill and R1-Preview on the GSM8K and MATH datasets suggest that both overly short and excessively long reasoning can degrade performance.

Shortest Correct Response Figure 7 shows the percentage of questions for which the shortest correct response appears at rank i. We find that for over 60% of questions—across all model and dataset combinations—the shortest sampled response is already correct. Among questions that the model is capable of answering (i.e., those with at least one correct response), the correct answer is often found within the first few shortest responses. Additionally, we analyze other rank related statistics, including the rank of the longest correct response, the shortest and longest incorrect responses, and the correlation between reasoning length and correctness. Due to space constraints, we refer interested readers to Appendix B for these extended results.

The findings from our sample-level analysis suggest that correct responses are often concentrated among the shorter or moderately long samples—where "short" and "moderate" are defined relative to the overall distribution of response lengths for a fixed question and model. We conjecture that this pattern arises because, once the i-th shortest response is already correct, generating longer responses beyond that point amounts to **overthinking**, i.e., it increases the risk of introducing errors or irrelevant information and compounding small mistakes, which can ultimately derail the reasoning process and reduce the likelihood of producing a correct answer.

5 Question-Level Analysis

At the question level, the relationship between response length and answer correctness becomes even more pronounced. We begin by comparing the average lengths of correct and incorrect responses. As shown in Table 1, incorrect responses are significantly longer than correct ones across all four model–dataset combinations. This trend is especially striking on the MATH dataset, where the average length of incorrect responses exceeds 6,000 tokens, while correct responses average fewer than 3,000 tokens for both models. In addition, we observe consistently strong negative Pearson and Spearman correlations between response length and accuracy—particularly on the MATH dataset—suggesting that longer responses are generally associated with lower correctness.

5.1 Impact of Question Difficulty on Response Length

To better understand why incorrect responses are significantly longer in the question level, we categorize questions based on model-specific difficulty levels. Specifically, we define three difficulty

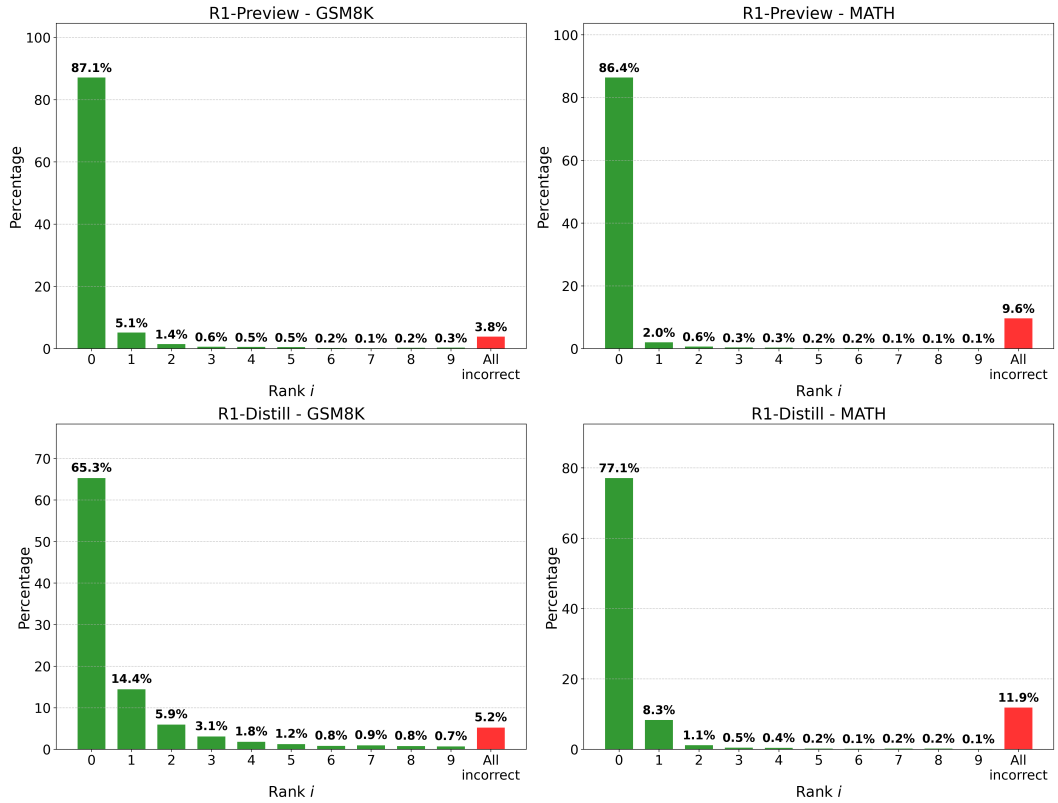

Figure 2: Percentage of questions for which the shortest correct response occurs at rank i.

Table 1: Correlation between token length and accuracy across four model-dataset combinations. We also report the average token length for correct responses ($L^{\checkmark}$) and incorrect responses ($L^{\times}$).

Model	Dataset	Pearson Correlation	Spearman Correlation	$L^{\checkmark}$ (Correct) (Mean ± Std)	$L^{\times}$ (Incorrect) (Mean ± Std)
R1-Preview	GSM8K	-0.5347	-0.3740	1374.82 ± 92.34	3069.22 ± 337.53
	MATH	-0.6835	-0.6390	2459.84 ± 29.92	6106.17 ± 37.70
R1-Distill	GSM8K	-0.1626	-0.2216	666.68 ± 131.59	842.46 ± 187.08
	MATH	-0.7248	-0.6801	2907.23 ± 275.25	6521.31 ± 433.08

tiers: *Easy*, *Medium*, and *Hard*. A question is labeled *Easy* if the model answers it correctly in all $N = 10$ sampled responses, *Hard* if it fails to answer it correctly in all 10 responses, and *Medium* if correctness varies across sampled responses. The number of questions in each category and their corresponding average response lengths are shown in Table 4 and Figure 11 respectively (see Appendix C). Consistent with the trends observed in Table 1, we find that *Easy* questions—those consistently answered correctly by the model—tend to have the shortest responses, while *Hard* questions—those consistently answered incorrectly—produce the longest. However, it remains unclear whether the increased length reflects the model's ability to recognize the intrinsic complexity of the question and adapt its reasoning accordingly, or whether the longer responses themselves introduce additional errors, making the question appear more difficult than it truly is. To disentangle these factors, we start by a cross-model analysis that examines response lengths within the subset of questions labeled as *Easy* for a given model.

Specifically, for two models $\mathcal{M}_i$ and $\mathcal{M}_j$, we compare generation lengths on three sets of questions: (1) the shared-easy set $Q_{\cap}^{\text{easy}} = Q_i^{\text{easy}} \cap Q_j^{\text{easy}}$, which both models find easy; (2) the $\mathcal{M}_i$-advantage set, $Q_{i\setminus j}^{\text{easy}} = Q_i^{\text{easy}} \setminus Q_j^{\text{easy}}$, which is easy for $\mathcal{M}_i$ but not for $\mathcal{M}_j$; and (3) the $\mathcal{M}_j$-advantage set, $Q_{j\setminus i}^{\text{easy}} = Q_j^{\text{easy}} \setminus Q_i^{\text{easy}}$, which is easy for $\mathcal{M}_j$ but not for $\mathcal{M}_i$. If model $\mathcal{M}_i$ generates longer responses on the $\mathcal{M}_i$-advantage set than on the shared-easy set, it suggests that questions in the $\mathcal{M}_i$-advantage

set are intrinsically harder for $\mathcal{M}_i$ than those in the shared-easy set—potentially requiring longer reasoning. A similar interpretation holds for model $\mathcal{M}_j$.

Table 2: T-test results comparing response lengths on the shared-easy set $Q_{\cap}^{\text{easy}}$ with $\mathcal{M}_1$-advantage set $Q_1^{\text{easy}} \backslash Q_2^{\text{easy}}$ and $\mathcal{M}_2$-advantage set $Q_2^{\text{easy}} \backslash Q_1^{\text{easy}}$ on $\mathcal{M}_1$ (R1-Preview) and $\mathcal{M}_2$ (R1-Distill), respectively. Significance is determined at $p < 0.05$.

		GSM8K		**MATH**	
		$\mathcal{M}_1$-Adv Set	**$\mathcal{M}_2$-Adv Set**	**$\mathcal{M}_1$-Adv Set**	**$\mathcal{M}_2$-Adv Set**
$\mathcal{M}_1$ (R1-Preview)	**Avg. Tokens on $Q_{\cap}^{\text{easy}}$**	1015.37		1690.62	
	Avg. Tokens $Q_{i \backslash j}^{\text{easy}}$	1348.60	1280.41	2832.86	2961.75
	T-Statistic	17.7067	7.1635	41.5385	14.7609
	P-Value	5.16×10^{-68}	1.08×10^{-12}	$0.00 \times 10^{+00}$	7.09×10^{-48}
	Significant?	Yes	Yes	Yes	Yes
$\mathcal{M}_2$ (R1-Distill)	**Avg. Tokens on $Q_{\cap}^{\text{easy}}$**	596.78		2322.63	
	Avg. Tokens $Q_{i \backslash j}^{\text{easy}}$	632.46	555.54	4012.31	3126.20
	T-Statistic	2.1656	-1.3379	37.9297	6.2503
	P-Value	3.04×10^{-2}	1.81×10^{-1}	4.87×10^{-279}	4.59×10^{-10}
	Significant?	Yes	No	Yes	Yes

Length Signals Perceived Difficulty in Easy Questions Table 2 reports the average token lengths and t-test statistics comparing these sets. We observe that for both models, the average response length increases on their respective advantage sets compared to the shared-easy set. For R1-Preview, the average token length increases from 1015.37 on the shared-easy set to 1348.60 on its own advantage set for GSM8K, and from 1690.62 to 2832.86 for MATH. This suggests that although R1-Preview consistently answers questions in its advantage set correctly, these questions may inherently require longer responses than those in the shared-easy set. Similarly, for R1-Distill, the average response length on MATH increases from 2,322.63 to 3,126.20 when comparing shared-easy to advantage-set questions. In contrast, while R1-Distill's response length decreases from 596.78 to 555.54 on its advantage set for GSM8K, this reduction is not statistically significant based on the p-value. Overall, these patterns indicate that both models generally treat their advantage-set questions as more difficult than those in the shared-easy set, as reflected in the increased length of their responses. Furthermore, we observe that both R1-Distill and R1-Preview's generation length has increased in another model's advantage set. This suggests that even when a model is unable to consistently answer such questions correctly, it may still recognize their increased difficulty and attempt to adapt by generating longer responses.

Table 3: T-test results comparing response lengths on shared-sard set $Q_{\cap}^{\text{hard}}$ with $\mathcal{M}_2$-Advantage set $Q_1^{\text{hard}} \backslash Q_2^{\text{hard}}$ and $\mathcal{M}_1$-Advantage set $Q_2^{\text{hard}} \backslash Q_1^{\text{hard}}$ on $\mathcal{M}_1$ (R1-Preview) and $\mathcal{M}_2$ (R1-Distill), respectively. Significance is determined at $p < 0.05$.

		GSM8K		**MATH**	
		$\mathcal{M}_2$-Adv Set	**$\mathcal{M}_1$-Adv Set**	**$\mathcal{M}_2$-Adv Set**	**$\mathcal{M}_1$-Adv Set**
$\mathcal{M}_1$ (R1-preview)	**Avg. Tokens on $Q_{\cap}^{\text{hard}}$**	2751.74		5845.98	
	Avg. Tokens $Q_{i \backslash j}^{\text{hard}}$	3826.86	3076.68	5240.06	6646.30
	T-Statistic	4.3064	1.6509	-2.2531	5.1719
	P-Value	2.29×10^{-5}	9.96×10^{-2}	2.46×10^{-2}	2.87×10^{-7}
	Significant?	Yes	No	Yes	Yes
$\mathcal{M}_2$ (R1-Distill)	**Avg. Tokens on $Q_{\cap}^{\text{hard}}$**	1253.09		6499.55	
	Avg. Tokens $Q_{i \backslash j}^{\text{hard}}$	1492.40	1001.46	5411.75	7386.94
	T-Statistic	1.2341	-1.7563	-4.3546	6.1999
	P-Value	2.18×10^{-1}	7.98×10^{-2}	1.53×10^{-5}	8.65×10^{-10}
	Significant?	No	No	Yes	Yes